

# Run three machine cycles

Name \_\_\_\_\_

**Your job:** run three instructions. Each one goes through the same four stages — **FETCH** → **DECODE** → **EXECUTE** → **WRITE**. The instructions are shortened codes, so you have to look each one up in the table to find out what it means.

| MEMORY             | STARTING STATE                          | DECODE LOOKUP TABLE                                                                                                                                                                                                                                                                                                                                                                                                                                |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
|--------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------|-------------|-------|-----------------------------------|---|-------|----------------------------------|---|-------|-----------------------------------------|---|-------|-----------------------------------------|------|
| 0: LDI 5           | PC = 0                                  | <table border="1"><thead><tr><th>Code</th><th>What it means</th><th>Destination</th></tr></thead><tbody><tr><td>LDI n</td><td>Load the number n into register A</td><td>A</td></tr><tr><td>ADI n</td><td>Add the number n to what is in A</td><td>A</td></tr><tr><td>SBI n</td><td>Subtract the number n from what is in A</td><td>A</td></tr><tr><td>STA n</td><td>Copy what is in A into memory address n</td><td>M[n]</td></tr></tbody></table> | Code | What it means | Destination | LDI n | Load the number n into register A | A | ADI n | Add the number n to what is in A | A | SBI n | Subtract the number n from what is in A | A | STA n | Copy what is in A into memory address n | M[n] |
| Code               | What it means                           | Destination                                                                                                                                                                                                                                                                                                                                                                                                                                        |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| LDI n              | Load the number n into register A       | A                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| ADI n              | Add the number n to what is in A        | A                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| SBI n              | Subtract the number n from what is in A | A                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| STA n              | Copy what is in A into memory address n | M[n]                                                                                                                                                                                                                                                                                                                                                                                                                                               |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| 1: ADI 1           | IR = empty                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| 2: STA 20          | A = unset                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |
| 20: (data – empty) | M[20] = empty                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |      |               |             |       |                                   |   |       |                                  |   |       |                                         |   |       |                                         |      |

## The trace — one row per instruction, one column per stage

| Instruction                            | 1. FETCH<br>Copy the instruction at the PC into the IR, then move the PC on. | 2. DECODE<br>Look the code up. What does it mean, and where does the result go? | 3. EXECUTE<br>Work out the result. Nothing is stored yet. | 4. WRITE<br>Put the result in the destination you decoded. |
|----------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------|
| <b>Cycle 1</b><br>PC starts at 0       | IR holds<br>_____<br>PC is now<br>_____                                      | Means<br>_____<br>Destination<br>_____                                          | Result<br>_____                                           | A =<br>_____                                               |
| <b>Cycle 2</b><br>use the PC you wrote | IR holds<br>_____<br>PC is now<br>_____                                      | Means<br>_____<br>Destination<br>_____                                          | Calculation<br>_____<br>Result<br>_____                   | A =<br>_____                                               |
| <b>Cycle 3</b><br>use the PC you wrote | IR holds<br>_____<br>PC is now<br>_____                                      | Means<br>_____<br>Destination<br>_____                                          | Value to be copied<br>_____<br>_____                      | Destination and value<br>_____<br>_____                    |

Stuck on a box? Every box has its own hint on the Canvas page for this activity.

### Final check

Ending state: PC = \_\_\_\_\_ A = \_\_\_\_\_ M[20] = \_\_\_\_\_

You ran **three** instructions. How many stages did each one pass through? \_\_\_\_\_

In cycle 3 the result did not go into register A. Which stage told you that?  
\_\_\_\_\_

Why could the machine not skip the decode stage on this worksheet?  
\_\_\_\_\_

**Model boundary:** a real machine stores instructions as numbers, not as short codes like **LDI**, and its lookup table is built into the hardware rather than printed on a page. We are hiding that encoding so the repeating cycle stays visible.